

ChatGPT

Centrality in Graph Theory

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explain centrality in graph

In graph theory and network analysis, **centrality** measures how “important” or “influential” a node is within a graph.

A graph can represent:

- people in a social network,
- computers in a network,
- webpages on the internet,
- airports connected by flights,
- neurons in the brain,
- or users/devices in IoT systems.

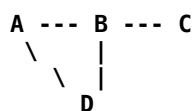
Different centrality measures define “importance” differently.

Basic Graph Terminology

A graph has:

- **Nodes (vertices)** → entities
- **Edges** → connections between entities

Example:



Nodes = A, B, C, D

Edges:

- A-B
- B-C
- A-D
- B-D

Why Centrality Matters

Centrality helps answer questions like:

- Which person spreads information fastest?

- Which router failure would disrupt the network most?
- Which webpage is most important?
- Which airport is the main hub?
- Which protein is most influential in a biological network?

Major Types of Centrality

The most common centrality measures are:

1. Degree Centrality
2. Closeness Centrality
3. Betweenness Centrality
4. Eigenvector Centrality
5. PageRank
6. Katz Centrality

1. Degree Centrality

Measures:

“How many direct connections does a node have?”

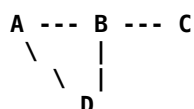
Formula:

$$C_D(v) = \deg(v)$$

Where:

- $\deg(v)$ = number of edges connected to node v

Example



Connections:

- $A \rightarrow 2$
- $B \rightarrow 3$
- $C \rightarrow 1$
- $D \rightarrow 2$

So:

- B has highest degree centrality.
-

Interpretation

High degree centrality means:

- many direct neighbors,
 - high local influence,
 - fast local communication.
-

Applications

- Social media influencers
 - Popular websites
 - Highly connected IoT devices
-

Limitation

A node may have many neighbors but still not be globally important.

2. Closeness Centrality

Measures:

“How close is a node to all other nodes?”

A node is important if it can quickly reach everyone else.

Formula:

$$C_C(v) = \frac{1}{\sum_u d(v,u)}$$

Where:

- $d(v, u)$ = shortest path distance
-

Idea

Smaller total distance → higher closeness centrality.

Example

A --- B --- C --- D

Node B:

- distance to A = 1
 - to C = 1
 - to D = 2
- Total = 4

Node A:

- to B = 1
 - to C = 2
 - to D = 3
- Total = 6

So B is more central.

Applications

- Fast information spreading
- Communication networks
- Emergency routing

Limitation

Does not work well in disconnected graphs.

3. Betweenness Centrality

Measures:

“How often does a node lie on shortest paths between other nodes?”

It identifies **bridge nodes** or **brokers**.

Formula:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where:

- σ_{st} = total shortest paths from s to t

- $\sigma_{st}(v)$ = paths passing through v

Example

A --- B --- C --- D

B and C are bridges.

If B fails:

- A becomes isolated.

So B has high betweenness.

Applications

- Detecting critical routers
- Influential mediators
- Traffic bottlenecks
- Disease spread control

Limitation

Computationally expensive for large graphs.

4. Eigenvector Centrality

Measures:

“A node is important if it connects to important nodes.”

Not all connections are equal.

Idea

Connecting to celebrities is better than connecting to random people.

Formula:

$$Ax = \lambda x$$

Where:

- A = adjacency matrix

- x = centrality vector

Interpretation

A node gets high score if:

- it has many neighbors,
- AND those neighbors are also influential.

Applications

- Social influence analysis
- Scientific citation networks
- Recommendation systems

5. PageRank Centrality

Used by Google search.

Improved version of eigenvector centrality.

Idea:

A webpage is important if important pages link to it.

Simplified Formula

$$PR(v) = \frac{1-d}{N} + d \sum_{u \in B_v} \frac{PR(u)}{L(u)}$$

Where:

- d = damping factor
- B_v = backlinks
- $L(u)$ = outgoing links

Applications

- Search engines
- Recommendation systems
- Citation ranking

6. Katz Centrality

Extension of eigenvector centrality.

Counts:

- direct neighbors,
- indirect neighbors too.

Farther nodes contribute less.

Formula:

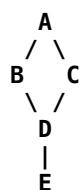
$$x = \alpha Ax + \beta$$

Comparison of Centrality Measures

Centrality	Main Idea	Best For
Degree	Number of neighbors	Local popularity
Closeness	Average distance	Fast communication
Betweenness	Bridge nodes	Bottleneck detection
Eigenvector	Connected to important nodes	Influence
PageRank	Link importance	Web ranking
Katz	Direct + indirect influence	Network propagation

Visualization Example

Consider:



- D has high betweenness (bridge to E)
- A/B/C have moderate degree
- D has strong closeness
- Nodes connected to strong nodes gain higher eigenvector score

Centrality in Different Networks

Social Networks

- Influencers
- Community leaders

Computer Networks

- Critical routers
- Attack targets

Transportation

- Airport hubs
- Traffic bottlenecks

Biology

- Important proteins/genes

IoT Networks

- Gateway nodes
- Efficient routing

Centrality in Directed Graphs

For directed graphs:

- In-degree centrality → incoming edges
- Out-degree centrality → outgoing edges

Example:

- Twitter followers → in-degree
- Following many people → out-degree

Weighted Centrality

If edges have weights:

- stronger relationships contribute more,
- shortest paths consider weights.

Example:

- flight frequency,
- bandwidth,
- trust score.

Computational Complexity

Measure	Complexity
Degree	Very low
Closeness	Medium
Betweenness	High
Eigenvector/PageRank	Iterative/high

Intuition Summary

Think of a city road network:

- **Degree** → busiest intersection
- **Closeness** → most centrally located place
- **Betweenness** → bridge/highway connector
- **Eigenvector** → connected to VIP locations
- **PageRank** → reputation passed through links

Real-World Example: Social Media

Suppose:

- User A has 1M random followers
- User B has 100K followers, but all are celebrities

Eigenvector/PageRank may rank B higher because influential neighbors matter.

In Multilayer Networks

Since you were asking about multilayer graphs earlier:

Centrality becomes more complex because:

- nodes may exist in multiple layers,
- influence can spread across layers,

- interlayer edges matter.

Examples:

- multiplex PageRank,
- tensor-based eigenvector centrality,
- multilayer betweenness.

These often use:

- adjacency tensors,
- supra-adjacency matrices,
- tensor decompositions.